

1. (Question 1). Let X be a discrete random variable with the given probability mass function with $0 \leq \theta \leq 1$ as a parameter:

x	0	1	2	3
$P(X = x \theta)$	$2\theta/3$	$\theta/3$	$2(1 - \theta)/3$	$(1 - \theta)/3$

We have the following i.i.d. sample $\mathbf{x} = (3, 0, 2, 1, 3, 2, 1, 0, 2, 1)$ from X and we want the maximum likelihood estimate of θ . Let x^t denote the t -th observation, for $t = 1, \dots, N$, where here $N = 10$, so that $\mathbf{x} = (x^1, \dots, x^N)$. The likelihood of the parameter θ is

$$\begin{aligned} l(\theta | \mathbf{x}) &= \prod_{t=1}^N p(x^t | \theta) \\ &= p(3 | \theta) \cdot p(0 | \theta) \cdot p(2 | \theta) \cdot p(1 | \theta) \cdot p(3 | \theta) \\ &\quad \cdot p(2 | \theta) \cdot p(1 | \theta) \cdot p(0 | \theta) \cdot p(2 | \theta) \cdot p(1 | \theta) \\ &= \frac{1 - \theta}{3} \cdot \frac{2\theta}{3} \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \cdot \frac{1 - \theta}{3} \\ &\quad \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \cdot \frac{2\theta}{3} \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \\ &= \left(\frac{2\theta}{3}\right)^2 \left(\frac{\theta}{3}\right)^3 \left(\frac{2(1 - \theta)}{3}\right)^3 \left(\frac{1 - \theta}{3}\right)^2 \\ &= \frac{2^5}{3^{10}} \theta^5 (1 - \theta)^5. \end{aligned}$$

We can optimize via the log likelihood and our desired MLE for θ is $\hat{\theta} = \arg \max_{\theta} (L(\theta | \mathbf{x}))$ so consider

$$L(\theta | \mathbf{x}) = \log(l(\theta | \mathbf{x})) = \log \left(\frac{2^5}{3^{10}} \theta^5 (1 - \theta)^5 \right)$$

Applying rules of logarithms we have

$$L(\theta | \mathbf{x}) = \log \frac{2^5}{3^{10}} + 5 \log \theta + 5 \log(1 - \theta)$$

Taking the derivative of L with respect to θ

$$\frac{dL}{d\theta} = 0 + 5 \cdot \frac{1}{\theta} + 5 \cdot \frac{1}{1 - \theta} \cdot (-1)$$

$$= \frac{5}{\theta} - \frac{5}{1-\theta}$$

Set the derivative equal to zero and solve for θ :

$$0 = \frac{5}{\theta} - \frac{5}{1-\theta}$$

$$\frac{5}{1-\theta} = \frac{5}{\theta}$$

$$\theta = 1 - \theta$$

$$\theta = \frac{1}{2}$$

Therefore the maximum likelihood estimate of θ , is $\hat{\theta} = \frac{1}{2}$.

2. (*Question 2*).